

BRAEDEN MARCHANT

Mechatronics Engineering

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🌐 [Portfolio](#)

EDUCATION

McMaster University | 2022-2027

Bachelor of Mechatronics Engineering (Co-Op)

SKILLS

- **CAD:** Autocad, Solidworks, Inventor, Fusion360, CATIA V5, 3D Printing, KiCad
 - **Programming:** Python, C++, C, Matlab/Simulink, Verilog, STM32, ESP32, FPGA
 - **Automation:** Mitsubishi PLC, Siemens PLC, Safety PLC, HMI Design, SCADA
 - **Vision Systems:** Cognex, MVTec HALCON + Deep Learning, OpenCV, PyTorch
 - **Project Management:** Root Cause Analysis, SPC, Budget Tracking
 - **Certifications:** Working at Heights, First Aid, Confined Space
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WORK EXPERIENCE

Honda | May 2025 - August 2026

Electrical Engineering Intern (Paint 2 - CAST)

- Project lead on multiple \$100K projects upgrading PLC systems, gaining hands-on experience in PLC programming, HMI design, wiring, electrical drawings, and project management.
- Project lead on Cognex vision system, used to detect underbody sealer presence, preventing multiple production downtime scenarios
- Supported multiple AI vision systems using HALCON Deep Learning for automated model and jig detection
- Supported multiple safety projects installing light curtains, area scanners, and safety limit switches. Gained safety PLC experience with Mitsubishi and Siemens.
- Installed a variety of sensors and electrical equipment such as: VFDs, RFID readers, pressure sensors, laser sensors, PID controllers

Alectra Utilities | May - August 2024

Station Sustainment / Protection & Control

- Performed electrical diagnostic tests on equipment in numerous substations
- Installed and wired electrical equipment and devices in substations such as lighting arresters, circuit breakers, and monitoring equipment
- Built and wired transformer panels using various electrical components and tools

Canada's Wonderland | June - August 2023

Mechanical Technician

- Completed daily preventative maintenance on various rides and roller coasters through safety checks and restoring damaged parts
 - Gained experience with many power tools and machine shop equipment
 - Completed various repairs and built new parts such as pneumatic pistons, cylinders, wheels, and air brakes
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PROJECT/TEAM EXPERIENCE

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| • Cobot - Programming Lead 2025-2026
Python, OpenCV, RTDE, Wiring, Robot Teach | • Hexpad - Custom Macropad 2026
PCB Design, Soldering, CAD / 3D Printing |
| • MDL - Designathon 2024/2025
CAD, Product Design | • Custom Micro FPV Drone 2026 - Ongoing
Electronics, Soldering, RF, 3D Printing |
| • Krystal Klear Robotic Mirror Arm 2023
Arduino, Wiring, CAD / 3D Printing | • FRC - Engineering Team Lead 2019-2022
CAD, Wiring, Hand Tools |